

QUANG-HUY NGUYEN

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RESEARCH INTERESTS

My research integrates multi-omics and clinical data using statistical and AI methods for multimodal modeling, uncertainty quantification, agentic reasoning, and federated learning. I use these approaches to uncover disease mechanisms and advance precision risk prediction and stratification across diverse populations.

EDUCATION

Auburn University **Auburn, AL**
Ph.D. in Computer Science & Software Engineering *08/2023 – Present*

Auburn University **Auburn, AL**
M.S. in Computer Science & Software Engineering *08/2023 – 12/2026*

Dai Nam University **Hanoi, Vietnam**
B.Sc in Pharmacy (5 years program) *10/2015 – 08/2020*

RESEARCH EXPERIENCE

Department of Computer Science, Hanoi University of Science and Technology **Hanoi, Vietnam**
Visiting Graduate Student *12/2024 – 01/2025*

Advisor: Associate Professor Duc-Hau Le.

- Developed *PAG-Agent*, a web-based, LLM-driven platform integrating bulk and single-cell RNA-seq workflows, including differential expression, pathway enrichment, meta-analysis, literature-assisted interpretation, and hypothesis generation; designed a context-aware pathway-prioritization framework that incorporates experimental conditions and research objectives to identify biologically relevant pathways while reducing generic enrichment signals.

Department of Computer Science & Software Engineering, Auburn University **Auburn, AL**
Graduate Research Assistant *08/2023 – 08/2024*

Advisors: Research Assistant Professor Phi Hung Bya and Associate Professor Tin Nguyen.

- Conducted methodological reviews of LC-MS-based targeted and untargeted metabolomics and large language models in biomedicine; Dockerized 13 metabolomics software tools as reproducible services to improve accessibility and deployment.
- Led computational analyses of *Candidozyma auris*, including BLAST-based homology mapping and KEGG/GO enrichment to characterize pathways associated with stress adaptation and antifungal resistance; contributed to manuscript preparation, resulting in a co-first-author publication in *Nature Communications*.

School of Computer Science & Engineering, Thuyloi University **Hanoi, Vietnam**
Visiting Scientist *10/2021 – 07/2023*

Advisor: Associate Professor Duc-Hau Le.

- Developed *DrGA*, an R toolkit integrating cancer driver-gene analysis, gene-set enrichment, gene-clinical association, module-level analysis, and patient stratification.

School of Computer Science & Engineering, Thuyloi University **Hanoi, Vietnam**
Visiting Student *12/2017 – 07/2019*

Advisor: Associate Professor Duc-Hau Le.

- Replicated published computational studies, designed experiments, and implemented statistical and machine-learning algorithms in R for high-dimensional omics data.

INDUSTRY EXPERIENCE

Center for Biomedical Informatics, Vingroup Big Data Institute **Hanoi, Vietnam**
Research Scientist *10/2020 – 09/2021*

Managers: Associate Professor Duc-Hau Le and Dr. Nam Sy Vo.

- Developed *oCEM*, an R package for detecting overlapping co-expressed gene modules using ICA/IPCA, with a permutation-based procedure for selecting the optimal number of components.
- Designed and implemented a multi-omics analysis pipeline that identified novel prognostic subgroups of breast cancer (log-rank $P = 0.002$ versus $P = 0.598$ for PAM50); led manuscript preparation and writing.
- Co-mentored an undergraduate student on computational clustering and characterization of uveal melanoma patient subgroups.
- Developed *DeepSSC*, a semi-supervised deep-learning framework for multi-omics cancer subtype classification and biomarker discovery; co-mentored 2 undergraduate students on model development.

- Contributed to the design and planning of the VN1K variant-calling pipeline for population-scale genomic analysis of the Vietnamese population and to manuscript preparation; this work resulted in a publication in *Nature Communications*.
- Led the Vietnamese type 2 diabetes project focused on polygenic risk prediction and mechanism discovery; identified Beta-cell +PI (proinsulin) as the only mechanism significantly associated with T2D in Vietnamese individuals; co-drafted the manuscript.

Research Intern

03/2020 – 10/2020

Internship mentor: Associate Professor Duc-Hau Le.

- Developed and implemented an R-based multi-omics pipeline for systematic identification and functional analysis of cancer driver genes.

HONORS AND AWARDS

Next Gen Leadership Award

Auburn, AL

AGBT Precision Health

07/2026

- Selected as **one of 10 recipients worldwide** from a highly competitive applicant pool spanning academia, healthcare, and industry.
- Invited to deliver a two-minute awardee spotlight and present *PAG-Agent* at the 2026 AGBT Precision Health Meeting.
- Received a **\$1,000 full scholarship** covering travel, lodging, and registration.

Benchmarks Resource Grant Program

Auburn, AL

Kaggle Research Grant Program

07/2026

- Awarded a Benchmarks Resource Grant providing up to **\$50/day** and **\$500/month** in compute/API credits.

Gold Reviewer Award

Auburn, AL

ICML 2026

05/2026

- Recognized as a **Gold Reviewer**, ranking in the **top 25% of eligible reviewers** based on Area Chair ratings.

Google PhD Fellowship in Health Research (University-Level Nomination)

Auburn, AL

Auburn University

04/2026

- Selected as **one of three Ph.D. students** nominated university-wide by the Auburn University Graduate School for the 2026 Google PhD Fellowship in Health Research.

Certificate of Merit for Academic Excellence and Class Leadership

Hanoi, Vietnam

Dai Nam University

08/2020

- Ranked in the top 3% of the department's 2015 cohort (nearly 110 students).
- Served as class president for five consecutive years (2015–2020).

27th Mathematics Olympic for Students

Nha Trang, Vietnam

Vietnam Mathematical Association

04/2019

- Received an **honorable mention** in Linear Algebra among 640 undergraduate competitors representing 80 institutions nationwide.
- Represented my university as its first and only participant and became the first Vietnamese non-engineering student in the history of the competition.

PEER-REVIEWED PUBLICATIONS

†co-first author, *co-corresponding author

- Trang T. H. Tran, *et al.*, **Quang-Huy Nguyen**, *et al.*, Quan Nguyen*, Van H. Vu*, Nam S. Vo*. *VN1K is a pangenome-informed multi-omics and phenomics resource for the Vietnamese population*. **Nature Communications**, 17:6679. doi:10.1038/s41467-026-75375-0 2026
- Ching-Lan Chang†, Michael A. Mosh†, **Quang-Huy Nguyen†**, *et al.*, David Hess*, Edwin C. Oh*. *Wastewater intelligence predicts the emergence of clinically-relevant and drug-resistant *Candidozyma auris* at healthcare facilities*. **Nature Communications**, 17:5415. doi:10.1038/s41467-026-71960-5 2026
- Quang-Huy Nguyen**, Ha Nguyen, Edwin C. Oh, Tin Nguyen. *Current approaches and outstanding challenges of functional annotation of metabolites: a comprehensive review*. **Briefings in Bioinformatics**, 25(6):bbae498. doi:10.1093/bib/bbae498 2024
- Duc-Hau Le, **Quang-Huy Nguyen**, Lan T.M. Dao. *Editorial: Computational and Experimental Approaches in Exploring the Role of Genetics and Genomics in Multifactorial Diseases*. **Frontiers in Genetics**, 13:873069. doi:10.3389/fgene.2022.873069 2022

7. **Quang-Huy Nguyen**, Tin Nguyen, Duc-Hau Le. *DrGA: cancer driver gene analysis in a simpler manner*. **BMC Bioinformatics**, 23:86. doi:10.1186/s12859-022-04606-0 2022
6. **Quang-Huy Nguyen**, Tin Nguyen, Duc-Hau Le. *Identification and validation of a novel three hub long noncoding RNAs with m6A modification signature in low-grade gliomas*. **Frontiers in Molecular Biosciences**, 9:801931. doi:10.3389/fmolb.2022.801931 2022
5. **Quang-Huy Nguyen**, Duc-Hau Le. *oCEM: Automatic detection and analysis of overlapping co-expressed gene modules*. **BMC Genomics**, 23:39. doi:10.1186/s12864-021-08072-5 2022
4. Thi Hai Yen Nguyen†, Tin Nguyen, **Quang-Huy Nguyen†***, Duc-Hau Le*. *Re-identification of patient subgroups in Uveal Melanoma*. **Frontiers in Oncology**, 11:731548. doi:10.3389/fonc.2021.731548 2021
3. **Quang-Huy Nguyen**, Duc-Hau Le. *Similarity calculation, enrichment analysis, and ontology visualization of biomedical ontologies using UFO*. **Current Protocols**, 1:e115. doi:10.1002/cpz1.115 2021
2. **Quang-Huy Nguyen**, Duc-Hau Le. *Improving existing analysis pipeline to identify and analyze cancer driver genes using multi-omics data*. **Scientific Reports**, 10:20521. doi:10.1038/s41598-020-77318-1 2020
1. **Quang-Huy Nguyen**, Hung Nguyen, Tin Nguyen, Duc-Hau Le. *Multi-omics analysis detects novel prognostic subgroups of breast cancer*. **Frontiers in Genetics**, 11:574661. doi:10.3389/fgene.2020.574661 2020

PEER-REVIEWED CONFERENCES

1. Hung Nguyen, Bang Tran, Duc Tran, **Quang-Huy Nguyen**, Duc-Hau Le, Tin Nguyen. *Disease subtyping using community detection from consensus networks*. **2020 12th International Conference on Knowledge and Systems Engineering (KSE)**, Can Tho, Vietnam, IEEE, 2020; 318–323. Oral Presentation. doi:10.1109/KSE50997.2020.9287843

BOOK CHAPTERS

1. Zongliang Yue*, Zeru Zhang, **Quang-Huy Nguyen**, Thanh Nguyen, Zhandos Sembay, Hao Chen, Robert Welner, Jake Chen*. *AI-driven gene-sets, networks, pathways, and interactions analyses of multi-omics data*. **Progress in Molecular Biology and Translational Science**, Vol. 221, pp. 277–313, Elsevier. doi:10.1016/bs.pmbts.2026.01.023 2026

WORKS UNDER REVIEW AND PREPARATION

†co-first author, *co-corresponding author

5. **Quang-Huy Nguyen**, Jiaqi Wang*, Wei-Shinn Ku*. *scBatchProx: Federated-Inspired Refinement for Stable Cell-Type Discriminability under Heterogeneous Batch Compositions*. **arXiv**, 2602.00423. (Under review). doi:10.48550/arXiv.2602.00423 2026
4. **Quang-Huy Nguyen**, Zeru Zhang, Duc-Hau Le*, Jake Y. Chen*, Wei-Shinn Ku*, Hao Chen*, Zongliang Yue*. *PAG-Agent: a biologist-oriented research assistant for context-aware pathway-level analysis and interpretation*. **bioRxiv**, 2026.06.02.729674. (Under review; **Next Gen Leadership Award**; **Kaggle Benchmarks Resource Grant**). doi:10.64898/2026.06.02.729674 2026
3. Hoang Le†, Van-Minh Nguyen†, Duc-Hai Tran, Van-Anh Tran, Van-Nui Nguyen, **Quang-Huy Nguyen†***, Duc-Hau Le*. *A Semi-Supervised Deep Learning Framework for Multi-Omics Cancer Subtype Classification and Biomarker Discovery*. **bioRxiv**, 2022.01.13.476268. (Under review). doi:10.1101/2022.01.13.476268 2026
2. **Quang-Huy Nguyen**, Jiaqi Wang*, Wei-Shinn Ku*. *Conformalized Neural Networks for Federated Uncertainty Quantification under Dual Heterogeneity*. **arXiv**, 2602.23296. (Under review). doi:10.48550/arXiv.2602.23296 2026
1. **Quang-Huy Nguyen**, Jiaqi Wang*, Wei-Shinn Ku*. *FedeKD: Energy-Based Gating for Robust Federated Knowledge Distillation under Heterogeneous Settings*. **arXiv**, 2605.05553. (Under review). doi:10.48550/arXiv.2605.05553 2026

PRESENTATIONS AND TALKS

1. 2026 AGBT Precision Health Meeting (**Next Gen Leadership Award**), The Loews Coronado Bay Resort, San Diego, CA. *Topic: PAG-Agent: A biologist-oriented virtual research assistant for context-aware pathway analysis, prioritization, and interpretation*. 2026
2. 2024 MW CTR-IN Annual Meeting, The Westin Las Vegas Hotel & Spa, Las Vegas, NV. *Topic: A Systems-level Approach for Predicting Patient Survival using Multi-omics Data*. 2024

RESEARCH PROJECTS AND COMPETITIONS

nano-scFMs ( Repo)

Auburn, AL

Independent Open-Source Project

06/2026 – Present

- Built a collection of lightweight, faithful PyTorch reimplementations of single-cell foundation models to improve readability, reproducibility, and accessibility for AI biology research.
- Reimplemented model architectures, preprocessing pipelines, tokenization workflows, pretrained checkpoint loading, and embedding extraction for single-cell `.h5ad` data.

Cell Maps for AI (CM4AI) CodeFest, UAB

Birmingham, AL

NIH Bridge2AI Consortium

08/2025

- Collaborated in a team of 5 to create separate embeddings of yeast fitness data, proteomic mass spectrometry, and cellular imaging, then fuse them into a co-embedding to construct a with-yeast multiscale subcellular hierarchy that uncovered hidden biological processes in triple-negative breast cancer.
- Implemented the analysis pipeline based on the agreed workflow in Python.

The 1000 Vietnamese Genomes Project (VN1K)

Hanoi, Vietnam

Vingroup Joint Stock Company

2019 – 2024

PI: Associate Professor Duc-Hau Le and Dr. Nam Sy Vo.

GenomeAsia100K

Hanoi, Vietnam

NTU and others

2017 – 2022

PI: Dr. Stephan Schuster (Singapore side) and Associate Professor Duc-Hau Le (Vietnam side).

TEACHING EXPERIENCE

Department of Computer Science and Software Engineering, Auburn University

Auburn, AL

Graduate Teaching Assistant for COMP-1220 Python & COMP-1230 Matlab

08/2024 – Present

- Teaching Python and Matlab fundamentals to ≈ 60 undergraduates and guiding them through problem-solving exercises in lab sections.

PROFESSIONAL SERVICE

- **Conference Reviewer:** ICML 2026, NeurIPS 2026, ICDM 2026, AAAI 2027.
- **Journal Reviewer:** Frontiers in Genetics (2021), Frontiers in Cell and Developmental Biology (2022), Frontiers in Bioinformatics (2026), Frontiers in Artificial Intelligence (2026), Functional & Integrative Genomics (2026).